ASSUME: Agentic Discovery of Interpretable, Literature-Grounded Concepts for Images Embeddings

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Motivation. For many computer vision tasks, images are usually first featurized as embeddings and then trained on. However these vision encoders are black boxes, their latent space are not directly interpretable by human and therefore the decision of the models that are trained on these embeddings cannot be understood as well. On the other hand, Multimodal LLM (MLLM) has the ability to describe images in natural language, which is great for human but bad for models as they don't generate embedding.

ASSUME. Here I present **A**uto-**S**ynthesized **S**emantic **U**nified **M**orphological **E**ncoder. I use (MLLM) agents (Gemini 3.8-flash and Gemini 3.5-flash-lite) to build a fully unsupervised, label-free Concept Bottleneck Model (CBM) [1]. It generates the latent space from semantic concepts extracted from unlabelled images, check each concept against the scientific literature, and then generate a rubric so that an embedding agent can score any unseen images. Because each dimension is a visual concept in natural language, a researcher can read the embedding of an image directly, edit a concept value to search for images, and easily understand the latent space or audit decision of downstream models. I validate ASSUME on scanning electron microscopy (SEM) micrographs, but nothing in it is specific to SEM, and the same agentic pipeline can be pointed at any scientific image dataset.

Agent Setup. Proposer agent looks at a small batch of micrographs is asked to identify morphological concepts that separate them and generate definition and marking scheme on a 0 to 5 scale (0 = absent, 5 = dominant). Many of these proposer agents are run in parallel. Merger agent prunes concepts with near-constant scores and consolidate concept with high correlation after viewing micrographs where they co-occur. The verifier agent retrieves relevant papers for each concept through the Valency Bond paper search MCP to ground the latent space in the SEM domain, retaining intrinsic specimen morphology while pruning instrument or imaging artifact. The verifier provides up-to-date scientific knowledge so the concept latent agrees with human expert knowledge. During runtime, the embedding agent scores each micrograph against the finished rubrics producing a concept vector for that micrograph, which can be easily read by human or machine.

Results. I ran ASSUME on a public dataset of SEM micrographs [2], using 14,620 micrographs from 8 morphology classes. The 46-dimensional ASSUME embedding can cluster the class unsupervised, with a 5-fold k-NN class accuracy of 85.6% (scored on a subset) compared to 90.1% for DINOv3 (384-D SOTA Vision Foundation Model) [3]. Unlike DINOv3 whose latent is not easily interpretable, ASSUME's can be easily steered. We can edit a single axis like particulate areal density or periodic array regularity to retrieves micrographs of the same material class with sparser particles or disordered fibers. ASSUME's latent are also nearly orthogonal ($|r| \leq 0.15$). Since verifier only retains microstructural features, micrograph taken by different detector or have different contrast still get matched as closest neighbours.

Impact and future work. ASSUME produces a descriptor that is learned from data like a VFM embedding, but is read and edited like a handcrafted descriptor, complete with rubrics and literature citation for every latent. Because every axis is a named morphology measure, quantitative structure–property analysis can be easily done without any manual annotation. ASSUME enables explainable image interpretation through its semantic latent embeddings grounded in domain knowledge.

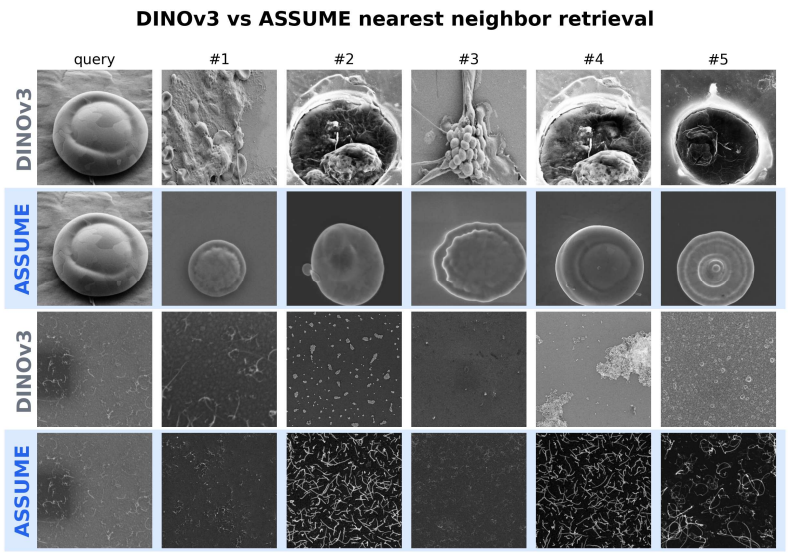


Figure 1: ASSUME retrieves microstructural similar micrographs while ignoring imaging conditions, while DINOv3 gets confused on microscopy specific concepts.

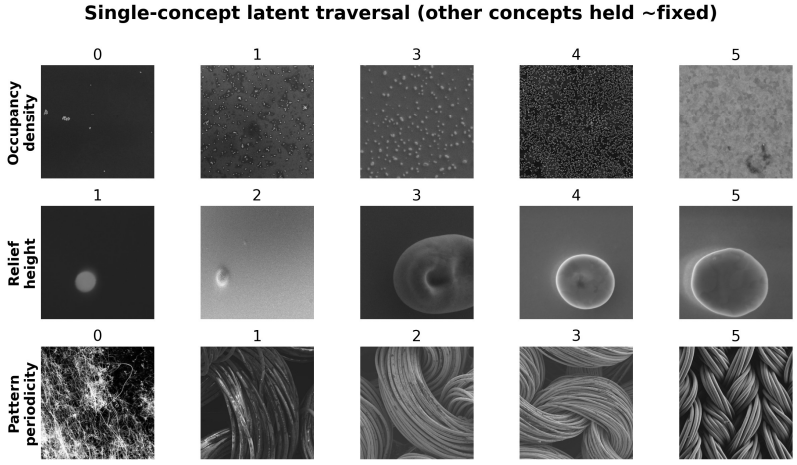


Figure 2: We can travel along the ASSUME latent to see how images changes. Top row shows all particles, only varies in density. Middle row shows just one particle, only varies in feature height. Bottom row shows fibers, only varies in periodicity.

References

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